

# NetSage AI

AI-Powered Intelligent Network Troubleshooting and Configuration Assurance  
Assistant

## 1. Background

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Modern networks are becoming increasingly complex due to the growth of cloud applications, distributed systems, IoT devices, and AI-driven workloads. Network administrators often face challenges in identifying the root cause of failures because issues can occur across multiple layers.

- Physical layer failures
- VLAN misconfigurations
- IP addressing problems
- Routing errors
- DHCP failures
- ACL/security rule mistakes
- Firewall restrictions

Traditional troubleshooting requires network engineers to manually inspect configurations, logs, and connectivity tests, which is time-consuming and requires expert knowledge. Cisco networks require intelligent systems that can assist engineers by automatically analyzing network information, identifying possible issues, and providing actionable recommendations.

## 2. Objective

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The objective of this project is to develop **NetSage AI**, an intelligent network troubleshooting assistant that combines:

- Network configuration analysis

- Python-based automation
- Artificial Intelligence
- Large Language Models (LLMs)
- Knowledge-based reasoning

The system should help students, beginners, and network administrators troubleshoot Cisco Packet Tracer-based networks by providing:

- Possible root cause
- Affected network layer
- Confidence score
- Recommended solution
- Verification steps

### 3. Proposed Solution

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NetSage AI will act as a **virtual network engineer**.

The user provides:

- Network topology information
- Device configurations
- Logs
- Error descriptions
- Connectivity test results

The system analyzes the information and generates an intelligent troubleshooting report.

### 4. System Workflow

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User Problem

|

| "Faculty cannot access server"

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## Network Data Collection

- Device Configs (VLAN details, IP addresses, ACL rules)
- Network Logs (Ping results, DHCP logs, Routing table)



## Python Analysis Engine



## AI Reasoning Layer (LLM + Network Knowledge Base)



## Final Diagnosis:

- Root Cause: VLAN configuration mismatch
- Network Layer: Layer 2
- Confidence: 92%
- Solution: Add VLAN to trunk port
- Verification: Run show vlan command

## 5. Key Features

### Feature 1: Network Configuration Analyzer

The system analyzes VLAN configuration.

#### Example Input:

```
Switch SW1
Configured VLANs:
- 10: Students
- 30: Guests

Expected:
- 10: Students
- 20: Faculty
- 30: Guests
```

#### AI Output:

- Issue Detected: Faculty VLAN 20 missing
- Impact: Faculty users cannot access academic resources
- Recommended Fix: Create VLAN 20 and allow it on trunk ports

### Feature 2: IP Address and DHCP Validation

The system checks incorrect subnet, duplicate IP addresses, wrong gateway, and DHCP mismatch.

### Example:

```
Student Device:
IP: 192.168.20.15
Gateway: 192.168.30.1
```

### AI Output:

- Problem: Gateway belongs to different subnet
- Layer: Layer 3
- Fix: Change gateway to 192.168.20.1

## Feature 3: Routing Problem Detection

The system identifies missing routes, wrong routing table entries, and gateway failures.

### Example:

```
Student Network
|
X
|
Server Network
```

### AI Analysis:

- Cause: No route exists between VLAN 10 and VLAN 40
- Solution: Configure inter-VLAN routing

## Feature 4: ACL Security Analysis

The system checks access rules.

### Example:

```
Existing ACL: DENY Faculty → Server
User complaint: "Faculty cannot access database"
```

### AI Analysis:

- Cause: ACL rule blocking faculty traffic

- Confidence: 95%
- Suggested action: Modify ACL rule

## 6. AI Components

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### 6.1 Knowledge Base

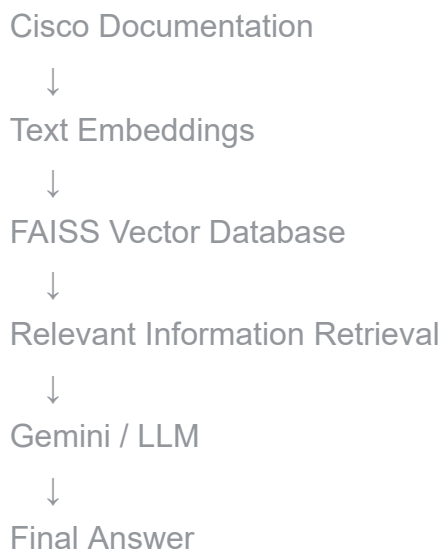
Create a Cisco networking knowledge repository containing:

- VLAN concepts
- Routing problems
- DHCP issues
- ACL rules
- Cisco commands
- Troubleshooting procedures

Storage: PDF documents, text files, and network manuals.

### 6.2 RAG-Based AI System

Instead of directly asking an LLM, use Retrieval Augmented Generation (RAG).



#### Advantages:

- More accurate answers
- Less hallucination
- Domain-specific responses

## 7. Network Simulation Environment

Use Cisco Packet Tracer to create a sample college network.

### Components:

- 1 Router
- 2 Switches
- 1 Wireless Access Point
- 1 Server
- 8 End Devices

| VLAN | Users      |
|------|------------|
| 10   | Students   |
| 20   | Faculty    |
| 30   | Guests     |
| 40   | Servers    |
| 50   | Management |

## 8. Fault Injection Module

To train/test the AI, intentionally introduce failures.

### Fault 1: Missing VLAN

- Change: Switch trunk allowed VLAN: 10, 30, 40 (Remove 20)
- Expected AI: Faculty connectivity failure - VLAN 20 missing from trunk
- Fix: Allow VLAN 20

### Fault 2: Wrong Gateway

- Change: Student Gateway from 192.168.20.1 to 192.168.30.1

- AI: IP configuration mismatch detected

### Fault 3: ACL Misconfiguration

- Block: Faculty → Server
- AI: Security policy preventing communication

## 9. Feedback Learning System

Human validation is important. After AI response, the user selects:

- Correct
- Partially Correct
- Wrong

Store feedback such as:

```
Problem: VLAN issue
AI Prediction: Correct
Confidence: 90%
```

This creates a dataset for future improvement.

## 10. Dashboard

Build a simple dashboard displaying network status and diagnosis history.

### Network Status:

- Devices: 8
- Active: 7
- Problems: 1

| Issue        | Prediction    | Confidence |
|--------------|---------------|------------|
| DHCP failure | Wrong gateway | 90%        |

| Issue       | Prediction | Confidence |
|-------------|------------|------------|
| No internet | NAT issue  | 85%        |

## 11. Technology Stack

| Component          | Technology                                                |
|--------------------|-----------------------------------------------------------|
| Frontend           | React / HTML-CSS-JS, Tailwind CSS                         |
| Backend            | Python FastAPI                                            |
| AI                 | Gemini API / OpenAI API, LangChain, Sentence Transformers |
| Vector Database    | FAISS                                                     |
| Data Processing    | Python, Pandas, JSON/YAML parsing                         |
| Network Simulation | Cisco Packet Tracer                                       |

## 12. Expected Final Output

A user enters:

```
"Guest users are unable to access internet"
```

NetSage AI responds:

```
Network Analysis Complete
```

```
Issue: Guest VLAN routing failure
```

```
Affected Layer: Layer 3
```

```
Root Cause: Missing NAT configuration for VLAN 30
```

```
Confidence: 89%
```

```
Recommended Fix: Configure NAT rule for guest subnet
```

```
Verification:
```

```
1. Ping gateway
```



2. Test internet connectivity
3. Check NAT translation table

## NetSage AI Project Documentation